

Fourth-order Difference Co-array based Unfolded Coprime Padded L-shaped Array Design

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ABSTRACT

This paper introduces an unfolded coprime padded L-shaped array (UCPLsA), which is constructed by augmenting the UCLsA with two additional subarrays to achieve a substantially larger number of uniform degrees-of-freedom (uDOF). The position distribution of the newly generated fourth-order virtual elements in UCPLsA is analyzed. When compared to current 2D sparse array designs, the proposed UCPLsA provides two major augmented proficiencies, offering a larger number of uDOF and effectively mitigating the mutual coupling effect. Numerical experiments are presented to corroborate the superior performance of UCPLsA in relation to alternative 2D sparse arrays.

Index Terms— Sparse Array, 2D DOA, Fourth-Order Difference Co-Array, Degree-of-Freedom.

1. INTRODUCTION

The emergence of sparse linear arrays has provided an effective solution to the challenge of achieving high-resolution and multi-target parameter estimation under the constraint of limited physical sensors [1–3]. Inspired by the one-dimensional (1D) sparse linear arrays [4–7], several two-dimensional (2D) sparse arrays have been proposed, such as billboard arrays, 2D nested arrays, and open box arrays [8–12], the partially open box array (POBA), half open box array (HOBA), half open box array with two layers (HOBA-2), and Hourglass arrays [13], which allows for analytical sensor-position expressions and achieves a favorable balance between hole-free co-array performance and lower mutual coupling.

In addition to DOA estimation based on second-order statistics, the fourth-order cumulants (FOCs) demonstrate higher robustness and resolution in complex interference and non-Gaussian environments. Compared with existing 1D arrays [14–16], the application of FOCs in 2D arrays remains

limited, primarily due to the complexity of constructing FOC, challenges in extending virtual array dimensions, and significantly increased computational cost. Among existing 2D arrays, the 2D super nested array (2D-SNA) [17] consists of two parallel super nested arrays with a hole-free FODCA. The augmented L-shaped nested array (AL-NA) [18] is constructed by combining two L-shaped nested subarrays, resulting in a virtual planar configuration that yields an extensive, hole-free uniform rectangular co-array.

In this study, we present a unfolded coprime L-shaped array (UCLsA)-based array structure and integrate it with the FODCA framework to enhance 2D DOA estimation performance. By extending the subarray placement of the UCLsA along the x - and y - axes, the resulting structure is able to generate a continuous virtual aperture in its fourth-order domain. The new 2D sparse geometry achieves a higher number of degrees of freedom which enables the detection of more sources compared with conventional 2D sparse arrays.

2. SIGNAL MODEL

The UCLsA [19] is constructed from two identical unfolded coprime arrays along the x - and y - axes. For an array of Q sensors, the corresponding sensor positions are represented by $\mathbb{P} \cdot d$, with $\mathbf{p}_i = (p_{xi}, p_{yi}) \in \mathbb{Z}^2$, where $i = 1, 2, \dots, Q$ denotes the integer coordinate of the i th sensor element, and d represents the inter-element spacing in units of wavelength, and λ stands for the signal wavelength. $\mathbf{p}_1 = (0, 0)$ is the reference point. Suppose a scenario with L narrowband far-field sources impinging on the array with DOAs as (η_l, ψ_l) , $l = 1, 2, \dots, L$, exhibiting non-Gaussian distributions and mutual independence. The parameters η_l and ψ_l represent the azimuth and elevation angles associated with the l th signal source. Accordingly, the signal received by the UCLsA is represented as

$$\mathbf{v}(t) = \sum_{l=1}^L \mathbf{m}(\mu_l, \chi_l) s_l(t) + \gamma(t) = \mathbf{M}\mathbf{s}(t) + \gamma(t), \quad (1)$$

where $\mathbf{s}(t) = [s_1(t), \dots, s_L(t)]^T$ is the signal vector with power σ_l^2 , $l = 1, \dots, L$. The noise vector $\gamma(t)$ is the vector of spatially white Gaussian noise. The steering matrix $\mathbf{M}$

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is given by $\mathbf{M} = [\mathbf{m}(\mu_1, \chi_1), \mathbf{m}(\mu_2, \chi_2), \dots, \mathbf{m}(\mu_L, \chi_L)]$, with

$$\mathbf{m}(\mu_l, \chi_l) = \exp\left(j \frac{2\pi}{\lambda} (p_x \mu_l + p_y \chi_l)\right), \quad (2)$$

where $\mu_l = \cos \alpha_l = \sin \eta_l \cos \psi_l$ and $\chi_l = \cos \beta_l = \sin \eta_l \sin \psi_l$.

3. ARRAY CONFIGURATION

After analysis, the aperture position expression of the fourth-order difference virtual array for the UCLsA can be expressed as $\mathbb{H}$, which is given in (3) (at the bottom of this page), where $\mathbb{H} = \mathbb{H}_2 \cup \mathbb{H}_3 \cup \mathbb{H}_4$, where $\smile$ stands for the exchange of Cartesian coordinates of the elements. $\langle n_1, n_2 \rangle$ denote the set of all integers in the integer interval $[n_1, n_2]$. And $\mathbb{X}_2 = \{k_1\varsigma - k_2v \mid k_1 \in \langle 1, v-1 \rangle, k_2 \in \langle 1, \varsigma-1 \rangle\}$.

Since the FODCA of the UCLsA contains numerous holes, in this section, two supplementary subarrays, $\mathbb{P}_3$ and $\mathbb{P}_4$, are incorporated into UCLsA, forming a new array structure named UCPLsA. The sensor positions of $\mathbb{P}_3$ and $\mathbb{P}_4$ are expressed as:

$$\begin{cases} \mathbb{P}_3 = \{(\varsigma + v, 0), (\varsigma + 2v, 0), (3\varsigma + v, 0)\}, \\ \mathbb{P}_4 = \{(0, \varsigma + v), (0, \varsigma + 2v), (0, 3\varsigma + v)\}. \end{cases} \quad (4)$$

Here, the sensor position set is denoted as $\mathbb{P}_{\text{UCPLsA}} = \mathbb{P}_{\text{UCLsA}} \cup \mathbb{P}_3 \cup \mathbb{P}_4$. The overall number of array elements in the UCPLsA is $D = Q + 3 + 3 = 2(\varsigma + v) + 3$. Fig.1 illustrates the physical array configuration of the UCPLsA for $\varsigma = 4$ and $v = 5$.

Definition 1 (Uniform DOF, uDOF): Given a 2D array $\mathbb{S}$, the total count of virtual array elements contained in the largest contiguous rectangular region which can be formed within its difference coarray.

Property 1: By introducing the subarrays $\mathbb{P}_3$ and $\mathbb{P}_4$, the additional virtual sensors in FODCA effectively eliminate the central missing lags in the virtual array of UCLsA. In the FODCA of the UCPLsA, the uDOFs region forms a rectangular area with side lengths $2L_x + 1$ and $2L_y + 1$, where $L_x = L_y = 4\varsigma + v - 1$, the cardinality of uDOFs is $T = (2L_x + 1) \times (2L_y + 1) = (8\varsigma + 2v - 1)^2$.

Proof: After introducing the subarrays $\mathbb{P}_3$ and $\mathbb{P}_4$, the additional virtual elements generated in the right-half plane of the second-order difference are represented as

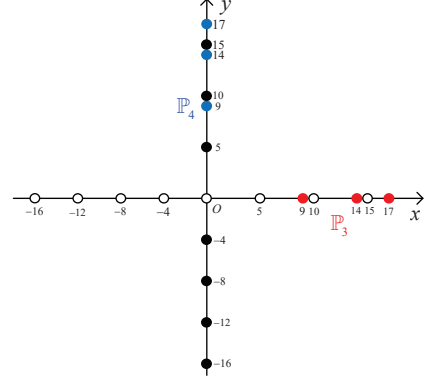


Fig. 1. Physical configuration of the UCPLsA for $\varsigma = 4$ and $v = 5$.

$$\begin{aligned} \mathbb{F}_e^{(2)} = \{ & (\mathbf{p}_{3X}, -\mathbf{p}_{3X}), (\mathbf{p}_{3X}, -b_1v), (a_1\varsigma, \mathbf{p}_{3X}), \\ & (\mathbf{p}_{3X}, a_1\varsigma), (b_1v, -\mathbf{p}_{3X}), (|\mathbf{p}_{3X} - \mathbf{p}_{3X}|, 0), \\ & (|\mathbf{p}_{3X} - b_1v|, 0), (\mathbf{p}_{3X} + a_1\varsigma, 0)\}. \end{aligned} \quad (5)$$

The second-order difference virtual elements originally located on the positive y -axis are represented as

$$\begin{aligned} \mathbb{F}_{e0}^{(2)} = \{ & (0, kv) (0, l\varsigma) (0, kv + l\varsigma) \mid \\ & k \in \langle 0, \varsigma - 1 \rangle, h_2 \in \langle 0, v - 1 \rangle\}. \end{aligned} \quad (6)$$

Case 1: The first set of additional fourth-order difference virtual sensors in the right-half plane $\mathbb{F}_{11}^{(4)}$, whose x -coordinates are $|a_1\varsigma - b_1v|$, is represented as

$$\begin{aligned} \mathbb{F}_{11}^{(4)} = \{ & (|a_1\varsigma - b_1v|, -(\mathbf{p}_{3X} + \mathbf{p}_{3Y})), \\ & (|a_1\varsigma - b_1v|, -b_1v - \mathbf{p}_{3X}), \\ & (|a_1\varsigma - b_1v|, \langle 0, v - 1 \rangle \varsigma - \mathbf{p}_{3X})\}, \end{aligned} \quad (7)$$

where $\mathbf{p}_{3X}$ denotes the x -coordinates of the sensors in $\mathbb{P}_3$. Let $\mathbf{f}_2 = \mathbf{p}_{3X} + \mathbf{p}_{3X}$, which can be written as

$$\begin{aligned} \mathbf{f}_2 = \{ & 2\varsigma + 3v, 2\varsigma + 2v, 2\varsigma + 4v, \\ & 4\varsigma + 2v, 4\varsigma + 3v, 6\varsigma + 2v\}. \end{aligned} \quad (8)$$

The second set of additional fourth-order difference virtual elements $\mathbb{F}_{12}^{(4)}$, whose x -coordinates are $|b_1v - \mathbf{p}_{3X}|$, can

$$\mathbb{H} = \begin{cases} \mathbb{H}_{21} = \{(\mathbb{X}_2, \pm(h_1\varsigma + h_2v)) \mid h_1 \in \langle 1, v-1 \rangle, h_2 \in \langle 1, \varsigma-1 \rangle\}, \\ \mathbb{H}_{31} = \{(qv + i\varsigma, \pm|a_1\varsigma - b_1v|)\} \cup \{(qv + i\varsigma, \pm\varsigma v)\}, \\ \mathbb{H}_{41} = \{(\langle \varsigma, 2\varsigma - 2 \rangle v, \pm(h_1\varsigma + h_2v)) \\ (\langle v + 1, 2v - 2 \rangle \varsigma, \pm(h_1\varsigma + h_2v)), (\langle v + 1, 2v - 2 \rangle \varsigma, \pm\langle v + 1, 2v - 1 \rangle \varsigma)\}, \\ \mathbb{H}_2 = \mathbb{H}_{21} \cup \widetilde{\mathbb{H}_{21}}, \quad \mathbb{H}_3 = \mathbb{H}_{31} \cup \widetilde{\mathbb{H}_{31}}, \quad \mathbb{H}_4 = \mathbb{H}_{41} \cup \widetilde{\mathbb{H}_{41}}. \end{cases} \quad (3)$$

be represented as $\mathbb{Y}_{12}$, which is given in (9) at the bottom of this page.

The x -coordinates of the third set $\mathbb{F}_{13}^{(4)}$ of fourth-order difference virtual elements $\mathbb{X}_{12}^{(4)}$ are given by

$$\begin{aligned}\mathbb{X}_{12}^{(4)} &= \{|\mathbf{p}_{3X} - \mathbf{p}_{3X}|\} \\ &= \pm\{\zeta, \zeta + v, 3\zeta, v - \zeta, 3\zeta - v, \\ &\quad 2v - \zeta, 3\zeta - 2v\}.\end{aligned}\quad (10)$$

While the y -coordinates in this case are the same as those in $\mathbb{F}_{12}^{(4)}$. It can be observed that $\mathbb{F}_{11}^{(4)}$, $\mathbb{F}_{12}^{(4)}$ and $\mathbb{F}_{13}^{(4)}$ can fill the holes in $\mathbb{H}_{21}$. After this, the y -coordinates of these virtual sensors are continuous over $[-4\zeta - v + 1, 4\zeta + v - 1]$.

Similarly, $\mathbb{F}_{11}^{(4)}$, $\mathbb{F}_{12}^{(4)}$, and $\mathbb{F}_{13}^{(4)}$ can fill the holes in $\mathbb{H}_{21}$.

Case 2: Since the structures of the x - and y - axes are identical, the newly added second-order virtual elements on the negative y -axis are given by $(0, -|\mathbf{p}_{3X} - \mathbf{p}_{3X}|)$, $(0, -|\mathbf{p}_{3X} - b_1v|)$, $(0, -|\mathbf{p}_{3X} - a_1\zeta|)$. The first set of additional fourth-order difference virtual elements $\mathbb{F}_{21}^{(4)}$, whose x -coordinates are $\mathbf{p}_{3X}$, is represented as

$$\mathbb{F}_{21}^{(4)} = \{(\mathbf{p}_{3X}, \mathbb{Y}_1)\}, \quad (11)$$

with $\mathbb{Y}_1 = \{v - \zeta, 2v - 2\zeta, 3v - 3\zeta, 2v - \zeta, 3v - 2\zeta\}$. The second set of additional fourth-order difference virtual elements also have the x -coordinates $\mathbf{p}_{3X}$, but with different y -coordinates, which can be expressed as

$$\begin{aligned}\mathbb{F}_{22}^{(4)} &= \left\{(\mathbf{p}_{3X}, \mathbb{Y}_{\mathbb{F}_{e0}^{(2)}}), (\mathbf{p}_{3X}, a_1\zeta + \mathbb{Y}_{\mathbb{F}_{e0}^{(2)}}), \right. \\ &\quad \left. (\mathbf{p}_{3X}, -b_1v - \mathbb{Y}_{\mathbb{F}_{e0}^{(2)}}), (\mathbf{p}_{3X}, -\mathbf{p}_{3X} - \mathbb{Y}_{\mathbb{F}_{e0}^{(2)}})\right\},\end{aligned}\quad (12)$$

where $\mathbb{Y}_{\mathbb{F}_{e0}^{(2)}}$ denotes the y -coordinates of $\mathbb{F}_{e0}^{(2)}$. The third set of additional fourth-order difference virtual elements $\mathbb{F}_{23}^{(4)}$, whose x -coordinates are $a_1\zeta + b_1v$, the corresponding y -coordinates can be represented as

$$\mathbb{Y}_{23} = \{\pm(a_1\zeta + \mathbf{p}_{3X}), b_1v \pm \mathbf{p}_{3X}, \mathbf{p}_{3X} - \mathbf{p}_{3X}\}. \quad (13)$$

In this case, the generated virtual elements can fill the central holes in $\mathbb{H}_{31}$ when $i = 1, q = 1$, $i = 1, q = 2$ and $i = 3, q = 1$, where i, q are the parameters of $\mathbb{H}_{31}$. After this, the corresponding virtual sensors exhibit continuity along the y -axis within $[-3\zeta v + 10v - 38, 2\zeta v - v]$. Similarly, the corresponding holes in $\mathbb{H}_{31}$ are also filled.

Case 3: When the x -coordinate is $b_1v + b_1v$, the y -coordinates of additional generated virtual elements $\mathbb{F}_{33}^{(4)}$ is

denoted as

$$\mathbb{Y}_{33} = \{(-b_1v - \mathbf{p}_{3X}), -a_1\zeta + \mathbf{p}_{3X}, \mathbf{p}_{3X} - \mathbf{p}_{3X}\}. \quad (14)$$

$\mathbb{F}_{33}^{(4)}$ can fill some holes in $\mathbb{H}_{41}$. When the x -axis is ζv , the vertical coordinates of the fourth-order differences after filling are continuous over the interval $[-4\zeta - v + 1, 4\zeta + v - 1]$. When the x -axis is $(\zeta + 1, 2\zeta - 2)v$, the vertical coordinates of the fourth-order differences after filling are continuous over the interval $[-\zeta v - 2\zeta - v, \zeta v + 2\zeta + v]$. Similarly, the corresponding holes in $\mathbb{H}_{41}$ are also filled. When the x -coordinate is $a_1\zeta + a_1\zeta$, the y -coordinates of additional generated virtual elements $\mathbb{F}_{34}^{(4)}$ are:

$$\mathbb{Y}_{34} = \{\pm(b_1v + \mathbf{p}_{3X}), -a_1\zeta + \mathbf{p}_{3X}, \mathbf{p}_{3X} + \mathbf{p}_{3X}\}. \quad (15)$$

$\mathbb{F}_{34}^{(4)}$ can fill the remaining holes in $\mathbb{H}_{41}$, and the y -coordinates are continuous over $[-4\zeta - v + 1, 4\zeta + v - 1]$.

The holes in $\mathbb{H}_{41}$ are also filled.

Case 4: When the x -coordinate is $\mathbf{p}_{3X} + b_1v$, the y -coordinates of additional generated virtual elements $\mathbb{Y}_{41}$ are:

$$\begin{aligned}\mathbb{Y}_{41} &= \{a_1\zeta - b_1v, a_1\zeta + a_1\zeta, -b_1v - \mathbf{p}_{3X}, \\ &\quad a_1\zeta - \mathbf{p}_{3X}, -\mathbf{p}_{3X} - \mathbf{p}_{3X}\}.\end{aligned}\quad (16)$$

When the x -coordinate is $\mathbf{p}_{3X} + a_1\zeta$, the y -coordinates of additional generated virtual elements are:

$$\begin{aligned}\mathbb{Y}_{42} &= \{a_1\zeta + b_1v, a_1\zeta - a_1\zeta, b_1v - \mathbf{p}_{3X}, \\ &\quad a_1\zeta + \mathbf{p}_{3X}, \mathbf{p}_{3X} - \mathbf{p}_{3X}\}.\end{aligned}\quad (17)$$

Combined with $\mathbb{F}_{23}^{(4)}$, the remaining holes in $\mathbb{H}_{31}$ can be filled, and the y -coordinates are continuous over $[-4\zeta - v + 1, 4\zeta + v - 1]$. Similarly, the remaining holes in $\mathbb{H}_{31}$ are also filled.

In summary, the central holes in the FODCA of UCLsA are effectively filled by introducing subarrays $\mathbb{P}_3$ and $\mathbb{P}_4$, with a total of $T = (2L_x + 1) \times (2L_y + 1) = (8\zeta + 2v - 1)^2$ uDOFs.

4. SIMULATION RESULTS

To demonstrate the proposed array's superiority in uDOFs, this section first compares UCPLsA with several typical 2D sparse arrays. For an identical count of physical sensors, the benchmark arrays include the Uniform Cross-shaped Array (UCsA), the 2D nested Array (2D-NA) [11, 12], HOBA [13]

$$\mathbb{Y}_{12} = \left\{ \begin{aligned} &((\langle -v + 2, v - 2 \rangle \zeta), \pm(a_1\zeta + b_1v), \pm(a_1\zeta + \mathbf{p}_{3X}), \\ &((\langle -\zeta + 2, \zeta - 2 \rangle v), \pm(b_1v - \mathbf{p}_{3X}), \mathbf{p}_{3X} - \mathbf{p}_{3X}. \end{aligned} \right. \quad (9)$$

and the generalized L-shaped array with odd-Even locations (GLA-OEL) [20]. The DOF comparison of these 2D sparse arrays are listed in Table I. As shown in Table I, the uDOFs of UCLsA are significantly increased from 13×13 to 41×41 after introducing the padded subarrays, achieving the highest value among all compared arrays. This confirms the effectiveness of the padding strategy in expanding the continuous virtual array.

Table I. Analysis of uDOFs across array structures

Array structure	Parameter	Physical sensor	uDOF
UCLsA	$\varsigma = 4, v = 5$	15	13×13
UCPLsA	$\varsigma = 4, v = 5$	21	41×41
UCsA	$\varsigma = 6$	21	21×21
GLA-OEL	$\varsigma = 7, v = 4$	21	36×21
2D-NA	$(N_{dx}, N_{dy}, N_{sx}, N_{sy}) = (3, 3, 3, 5)$	23	25×25
HOBA	$N_x = 7, N_y = 8$	22	25×29

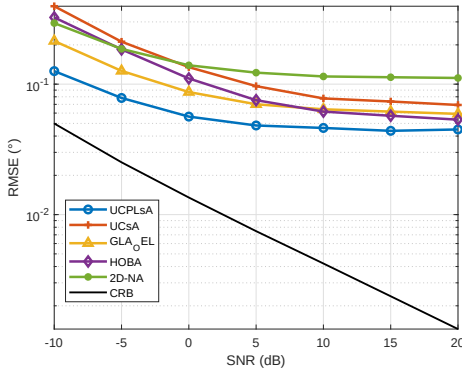


Fig. 2. RMSE versus SNR at $T = 5000$.

Under the condition without mutual coupling, Fig. 2 illustrates the Root Mean Square Error (RMSE) variation across various Signal-to-Noise Ratios (SNR) for 1000 independent Monte Carlo trials with snapshots $T = 5000$, two uncorrelated narrowband sources located in the far-field, $L = 2$, using the MUSIC algorithm. The Cramér-Rao bound (CRB) is also included as a standard reference [21, 22]. The SNR ranges from -15dB to 20dB . Among all compared arrays, UCPLsA maintains the minimum RMSE across the entire SNR range, with a particularly pronounced advantage in the low-SNR region, demonstrating superior estimation accuracy. The GLA-OEL has similar trend with UCPLsA, but its estimation accuracy is noticeably lower. 2D-NA exhibits relatively poor accuracy, although its performance remains stable. In contrast, the RMSE curves of UCsA and HOBA are relatively large in the low-SNR region and converge more slowly, indicating limited estimation stability under small-sample conditions.

Fig.3 shows the RMSE variation across different snapshot

settings for 1000 independent Monte Carlo trials at $\text{SNR} = 15\text{dB}$. The snapshots range from 800 to 6000. UCPLsA consistently achieves the lowest RMSE across all snapshots, and its error decreases faster than the other arrays as snapshot number increases. The estimation accuracy of GLA-OEL and 2D-NA is comparable, both outperforming UCsA and HOBA, indicating good adaptability to changes in different number of snapshots. UCsA and HOBA exhibit larger errors in the low-snapshot region, highlighting their limited estimation stability under small-sample conditions.

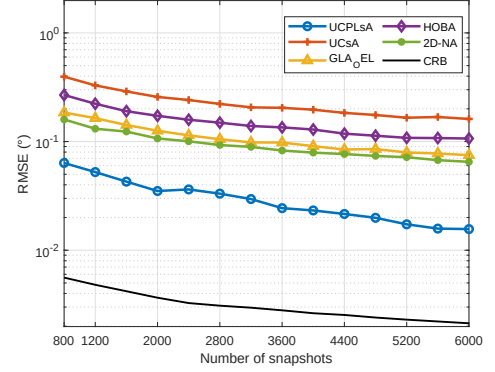


Fig. 3. RMSE versus snapshot number at $\text{SNR} = 15\text{dB}$.

5. CONCLUSION

This paper proposes a FODCA-based sparse array named UCPLsA, which is constructed by introducing additional subarrays into the UCLsA. Theoretical analysis shows that the addition of these subarrays generates new fourth-order virtual elements, which fill the central holes in UCLsA, forming a continuous FODCA in the central region and providing an increased number of uDOFs. Moreover, compared with existing 2D arrays, UCPLsA is capable of constructing a larger hole-free virtual array while maintaining relatively low mutual coupling effects. Simulation results demonstrate the effectiveness of the proposed UCPLsA.

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